T3 D_+2: full general UV Lagrangian and RGEs

Matchete/RGBeta symbolic output in mathematical notation

D_+2

Additional quartics: 0, additional bilinears: 0, extra Yukawa: none.
General matching and Wilson coefficients are not computed in this symbolic report.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Complete BSM UV Lagrangian

Matchete BSM Lagrangian, excluding the pure Standard Model contribution. $\mathcal{L}_{\text{BSM}} = \text{Coupling(MS1, \{\}, 0)}^2\text{Field(NewScalar1, Scalar, \{Index(d\$1,}$

$\text{+Coupling(lambdaT3, \{\}, 0)Bar(Field(NewScalar2, Scalar, \{\}, \{\}))Field(NewScalar1, Scalar, \{Index(d\$1, T3BSMd3)\}, \{\})Field(}H\text{, Scalar, \{Index(d\$1, SU2L(fund))\}, \{\})Field(}H\text{, Scalar, \{Index(d\$2, SU2L(fund))\}, \{\})CG(T3MixCG, \{Bar(Index(d\$1, SU2L(fund))), Bar(Index(d\$2, SU2L(fund))), Index(d\$1, T3BSMd3)\})$

$$+\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\text{Index}(\text{d\$\$1},\text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\text{Index}(\text{d\$\$1},\text{Lorentz})\}))$$

$$+\tfrac{1}{2}\text{Coupling}(\text{lambdaH1Inv1},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3HiggsPortal1CGInv1},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))),\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3})\})$$

$$+\tfrac{1}{2}\text{Coupling}(\text{lambdaH1Inv1},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3HiggsPortal1CGInv1},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))),\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})$$

$$+\tfrac{1}{2}\text{Coupling}(\text{lambdaH1Inv2},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3HiggsPortal1CGInv2},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))),\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})$$

$$+\tfrac{1}{2}\text{Coupling}(\text{lambdaH1Inv2},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3HiggsPortal1CGInv2},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))),\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})$$

$$+\tfrac{1}{2}\text{Coupling}(\text{lambdaS1Inv1},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$3},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$4},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\}))\text{CG}(\text{T3Scalar1SdfCGInv1},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$3},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$4},\text{T3BSMd3})\})$$

$$+\tfrac{1}{2}\text{Coupling}(\text{lambdaS1Inv2},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$3},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$4},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\}))\text{CG}(\text{T3Scalar1SdfCGInv2},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$3},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$4},\text{T3BSMd3})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3},\{\},0))\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{Bar}(\text{T3MixCG}),\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{MS2},\{\},0)^2\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\})(-\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\})))$$

$$+\text{Coupling}(\text{lambdaH2Inv1},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{CG}(\text{T3HiggsPortal2CGInv1},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund})))\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))$$

$$+\text{Coupling}(\text{lambda12Inv1},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\}))\text{CG}(\text{T3CrossScalarCGInv1},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3})\})\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS2Inv1},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\})^2\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\}))^2$$

$$+\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\text{Index}(\text{d\$\$1},\text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\text{Index}(\text{d\$\$1},\text{Lorentz})\}))$$

$$-\text{Coupling}(\text{MF},\{\},0)\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\})),\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\}))$$

$$+\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\}))\text{Coupling}(\text{y2},\{\text{Index}(\text{d\$\$1},\text{Flavor}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},0)\text{CG}(\text{T3Y2CG},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$1},\text{T3BSMd2}))\})\text{NCM}(\text{Bar}(\text{Field}(l,\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{Flavor})\},\{\})),\text{DiracProduct}(\text{Proj}(1)),\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\}))$$

$$+\text{Coupling}(\text{y1},\{\text{Index}(\text{d\$\$1},\text{Flavor}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})\text{CG}(\text{T3Y1CG},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(l,\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{Flavor})\},\{\})),\text{DiracProduct}(\text{GammaCC},\text{Proj}(1)),\text{Transp}(\text{Bar}(\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\}))))$$

$$+i\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\})),\text{DiracProduct}(\text{GammaM}(\text{Index}(\text{d\$\$1},\text{Lorentz}))),\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\text{Index}(\text{d\$\$1},\text{Lorentz})\}))$$

$$+\text{Field}(\text{NewScalar2},\text{Scalar},\{\},\{\})\text{Bar}(\text{Coupling}(\text{y2},\{\text{Index}(\text{d\$\$1},\text{Flavor}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},0))\text{CG}(\text{Bar}(\text{T3Y2CG}),\{\text{Bar}(\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))),\text{Index}(\text{d\$\$1},\text{T3BSMd2})\})\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\})),\text{DiracProduct}(\text{Proj}(-1)),\text{Field}(l,\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{Flavor})\},\{\}))$$

$$+\text{Bar}(\text{Coupling}(\text{y1},\{\text{Index}(\text{d\$\$1},\text{Flavor}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},0))\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{CG}(\text{Bar}(\text{T3Y1CG}),\{\text{Bar}(\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))),\text{Bar}(\text{Index}(\text{d\$\$1},\text{T3BSMd2})),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})\text{NCM}(\text{Transp}(\text{Field}(l,\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{Flavor})\},\{\})),\text{DiracProduct}(\text{GammaCC},\text{Proj}(-1)),\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\}))$$

Interaction inventory

- `Yukawa1_GenericCG`
- `Yukawa2_GenericCG`
- `Scalar1SelfInv1`
- `Scalar1SelfInv2`
- `Scalar2SelfInv1`
- `HiggsPortal1Inv1`
- `HiggsPortal1Inv2`
- `HiggsPortal2Inv1`
- `ScalarCrossPortalInv1`
- `T3ScalarMix_GenericCG`

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2\,\frac{dg_1}{d\ln\mu} = \frac{163{g_1}^3}{6}$$

g_2

$$16\pi^2\,\frac{dg_2}{d\ln\mu} = -\frac{{g_2}^3}{2}$$

g_3

$$16\pi^2\,\frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}16\pi^2\,\frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\ & + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\ & + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\ & - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\ & - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\ & - 8{g_3}^2\,(Y_u)_{ij} \\ & - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}\end{aligned}$$

Y_d

$$\begin{aligned}16\pi^2\,\frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\ & + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\ & + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\ & + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\ & - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\ & - 8{g_3}^2\,(Y_d)_{ij} \\ & - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} &= 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ &+ 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 2\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}) \\ &- \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ &- \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

y_1

$$\begin{aligned}16\pi^2 \frac{dy_1}{d\ln\mu} &= T_\nu^{(1)}\,(y_1)_{ij} \\ &+ \frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(y_1)_{ij} \\ &- \frac{15}{2}{g_1}^2\,(y_1)_{ij}\end{aligned}$$

y_2

$$\begin{aligned}16\pi^2 \frac{dy_2}{d\ln\mu} &= 2T_\nu^{(2)}\,(y_2)_{ij} \\ &+ \frac{3}{4}\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} \\ &+ \frac{1}{2}\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} \\ &+ \left(y_2\ (y_2)^\dagger\ y_2\right)_{ij} \\ &- \frac{9}{2}{g_2}^2\,(y_2)_{ij} \\ &- \frac{15}{2}{g_1}^2\,(y_2)_{ij}\end{aligned}$$

M_F

$$\begin{aligned}16\pi^2 \frac{dM_F}{d\ln\mu} &= \frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 2\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(M_F)_{ij} \\ &- \frac{27}{2}{g_1}^2\,(M_F)_{ij}\end{aligned}$$

m_1^2

$$16\pi^2 \frac{dm_1^2}{d\ln\mu} = m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$$

$$
\begin{aligned}
&- 4\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)\\
&- 12{g_2}^2 m_1^2\\
&- 6{g_1}^2 m_1^2\\
&+ 2\lambda_{12}^{(1)} m_2^2
\end{aligned}
$$

m_2^2

$$
\begin{aligned}
16\pi^2\,\frac{dm_2^2}{d\ln\mu} = m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})\\
&- 8\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)\\
&- 24{g_1}^2 m_2^2\\
&+ 6\lambda_{12}^{(1)} m_1^2
\end{aligned}
$$

λ_1

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_1}{d\ln\mu} = \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\
&- 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)\\
&- 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)\\
&+ 4\lambda_5\,(\lambda_5)^*\\
&+ \frac{9{g_2}^4}{4}\\
&+ {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right)\\
&- 9{g_2}^2\lambda_1\\
&+ \frac{3{g_1}^4}{4}\\
&+ 12{\lambda_1}^2\\
&+ 6{\lambda_{HS_1}^{(1)}}^2\\
&+ 3{\lambda_{HS_1}^{(A)}}^2\\
&+ 2{\lambda_{HS_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = \lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})\\
&- \frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)\\
&+ 22{g_2}^4\\
&+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_1}^{(1)}\right)\\
&- 24{g_2}^2\lambda_{S_1}^{(1)}\\
&+ 12{g_1}^4
\end{aligned}
$$

$$
\begin{aligned}
&+ 2{\lambda_{12}^{(1)}}^2 \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 14{\lambda_{S_1}^{(1)}}^2 \\
&+ 16{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = &\, 8\lambda_{S_2}^{(1)} T_\nu^{(2)} \\
&- 8\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&+ 192{g_1}^4 \\
&- 48{g_1}^2\lambda_{S_2}^{(1)} \\
&+ 6{\lambda_{12}^{(1)}}^2 \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 10{\lambda_{S_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = &\, \lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_1}^{(A)}) \\
&- 2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
&+ 4\lambda_5\,(\lambda_5)^* \\
&+ 6{g_2}^4 \\
&- \frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 3{g_1}^4 \\
&- \frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6{\lambda_{HS_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = &\, \lambda_{HS_2}^{(1)}(4T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&- 4\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
&+ 12\lambda_5\,(\lambda_5)^*
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{9{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 12{g_1}^4 \\
&- \frac{51{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = {} & \lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)}) \\
& - 4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& + 4\lambda_5\,(\lambda_5)^* \\
& - 12{g_2}^2\lambda_{12}^{(1)} \\
& + 48{g_1}^4 \\
& - 30{g_1}^2\lambda_{12}^{(1)} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + 8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
& + 4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} = {} & \lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) - 4\lambda_{HS_1}^{(A)}) \\
& - \frac{21{g_2}^2\lambda_5}{2} \\
& - \frac{33{g_1}^2\lambda_5}{2} \\
& + 2\lambda_{12}^{(1)}\lambda_5 \\
& + 2\lambda_1\lambda_5 \\
& + 4\lambda_{HS_1}^{(1)}\lambda_5 \\
& + 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = {} & \lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_1}^{(A)}) \\
& - \frac{4}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& - \frac{8}{3}\lambda_5\,(\lambda_5)^*
\end{aligned}
$$

$$
\begin{aligned}
&+ {g_1}^2 \left(4{g_2}^2 - \frac{15\lambda_{HS_1}^{(A)}}{2} \right) \\
&- \frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{S_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(A)}}{d\ln\mu} = {} & 4\lambda_{S_1}^{(A)} T_\nu^{(1)} \\
& - \frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger \Big) \\
& - 2{g_2}^4 \\
& + {g_1}^2 \left(8{g_2}^2 - 12\lambda_{S_1}^{(A)} \right) \\
& - 24{g_2}^2\lambda_{S_1}^{(A)} \\
& + 3{\lambda_{HS_1}^{(A)}}^2 \\
& + 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
& + 20{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$